

PHY301 assignment 1

LAST DATE 27 May 2022

ORANGE MONKEY TEAM

Follow these Instruction before submitted assignment

- **Write your ID on the top of your solution file.**
- **All students are directed to use the font and style of text as is used in this document.**
- **Don't use colorful back grounds in your solution files.**
- **Use Math Type or Equation Editor etc for mathematical symbols.**

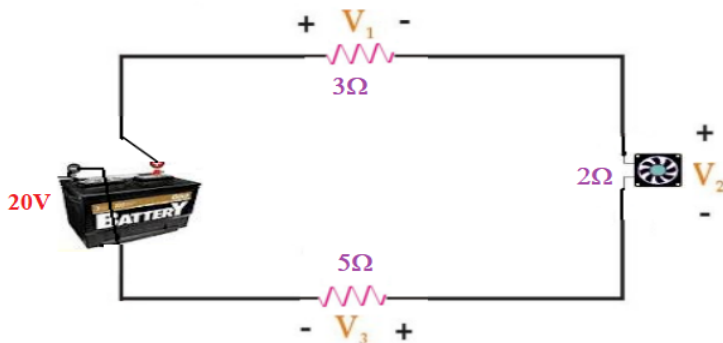
Question no 1 (PART A)

A 20V battery is attached to a circuit containing 2 resistors and a fan having 2Ω resistance.

Marks 10

- 1) Calculate the source current I_s .
- 2) Determine the voltages V_1 , V_2 , and V_3 .
- 3) Calculate the power dissipated by each resistor.

- 4) Determine the power delivered by the source, and compare it to the sum of the power levels of part (4)



1. Calculate the source current I_s

SOLUTION

To calculate the source current I_s, Firstly, we calculate the total resistance

$$R_T = R_1 + R_2 + R_3$$

$$R_T = 10\Omega$$

$$V = IR$$

AND

$$I_s = V_S / R_T$$

$$I_s = 20/10$$

$$I_s = 2A$$

2. Determine the voltages V₁, V₂, and V₃

SOLUTION

Since all resistances are in series, same 2A current pass through each resistance

$$V_1 = R_1 \cdot I_s$$

$$V_1 = 3 \cdot 2$$

$$V_1 = 6V$$

And

$$V_2 = R_1 \cdot I_s$$

$$V_2 = 2 \times 2$$

$$V_2 = 4V$$

And

$$V_3 = R_3 \times I_s$$

$$V_3 = 5 \times 2$$

$$V_3 = 10V$$

3. Calculate the power dissipated by each resistor

Solution

$$P_1 = I_2 \times R_1$$

$$P_1 = 2(2) \times 3$$

$$P_1 = 12W$$

and

$$P_2 = I_2 \times R_2$$

$$P_2 = 2(2) \times 2$$

$$P_2 = 8W$$

and

$$P_3 = I_2 \times R_3$$

$$P_3 = 2(2) \times 5$$

$$P_3 = 20W$$

4. Determine the power delivered by the source, and compare it to the sum of the power levels

Solution

$$P_s = V_s / I_s$$

$$P_s = (20)(2)$$

$$P_s = 40W$$

and

$$P_T = P_1 + P_2 + P_3$$

$$P_T = 12 + 8 + 20$$

$$P_T = 40W$$

Question. No: 1 PART B

Answer the following short questions.

Marks 5

- **Why a crow sitting on bare 100KV electric wire not get electric shock.**
Solution

- Because the crow is sitting on only one string, its body is not making its way to the other string of different abilities. Electrical current must flow in a circuit. To receive a shock, your body must complete a circuit between one point of the electrical potential and the other. So the reason we are shocked by touching a wire is that we often stand on a conductor.

JUST for undersatand don't mention translation in your assignment

چونکہ کوا صرف ایک تار پر بیٹھا ہوتا ہے، اس لیے اس کا جسم مختلف صلاحیتوں کی دوسری تار تک نہیں پہنچ پاتا۔ ایک سرکٹ میں برقی رو بہنا ضروری ہے۔ جھٹکا لینے کے لیے، آپ کے جسم کو برقی صلاحیت کے ایک نقطہ اور دوسرے کے درمیان ایک سرکٹ مکمل کرنا چاہیے۔ لہذا تار کو چھونے سے ہمیں چونکنے کی وجہ یہ ہے کہ ہم اکثر کنڈکٹر پر کھڑے ہوتے ہیں۔

- **Given below are some materials. Mention the conductor, semiconductor and Insulator from them.**

Silicon, Silver, iron, carbon, glass, copper, rubber.

- The semiconductor is Carbon and silicon
- The insulator is Glass

- The conductor is Copper , silver, Iron

